Predicting Total Medicare Beneficiaries by State (2023)

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Introduction

1. Is an interaction between average age and political affiliation linked to larger Medicare beneficiary totals?
2. Does greater health-care provider availability, measured by physicians per 100,000 residents, correspond to higher total Medicare beneficiary counts per state?
3. Does higher median family income correspond to lower total Medicare beneficiary counts per state?

Data Summary

This report only uses one 2023 observation for each of the 50 states. Before calculating the population quartiles, the total-state-population values for AK, AL, AR, and AZ were adjusted to be consistent with the listed state abbreviations provided in the original data set. The response variable, total Medicare beneficiaries, is strongly right-skewed: counts range between 120,280 and 7,080,940, with mean of 1,360,708, a median of 994,359, and IQR of 1,132,502. Because the mean is well above the median and the strong influence of a small number of extremely large states dominating the upper tail, a log transformation of the response variable seems appropriate before fitting the final regression model.

Among the quantitative predictors, HCC score has the clearest positive association with beneficiary totals, with Pearson correlation 0.562 on the untransformed response variable and 0.702 when using `log(Y)`. Percent male shows the strongest negative correlation of the quantitative predictors, with correlations -0.28 on the untransformed response variable and -0.467 when using `log(Y)`. Average age appears weak on its own (r = -0.038 on the raw scale), which makes the political affiliation-age interaction worth investigating rather than relying on a simple main-effect interpretation. Median income is nearly unrelated to the raw response variable (r = 0.051) and only very weakly negatively correlated when using `log(Y)` (r = -0.019), so any income effect may depend on adjustment for other predictors.

Appendix A: Data Dictionary

Data dictionary for the cleaned analysis dataset.

Variable	Type	Description
State	Identifier	Two-letter state abbreviation.
Beneficiaries	Response	Total Medicare beneficiaries in the state.
Avg_Age	Quantitative	Average beneficiary age.
HCC_Score	Quantitative	Hierarchical condition category risk score.
Pct_Male	Quantitative	Proportion of beneficiaries who are male.
Median_Income	Quantitative	Median annual family income.
Political_Affiliation	Qualitative	State political affiliation from the source file.
Education_Q	Qualitative	Quartile of bachelor's degree percentage by state.
Physician_Q	Qualitative	Quartile of physicians per 100,000 residents.
Population_Q	Qualitative	Quartile of corrected total state population.

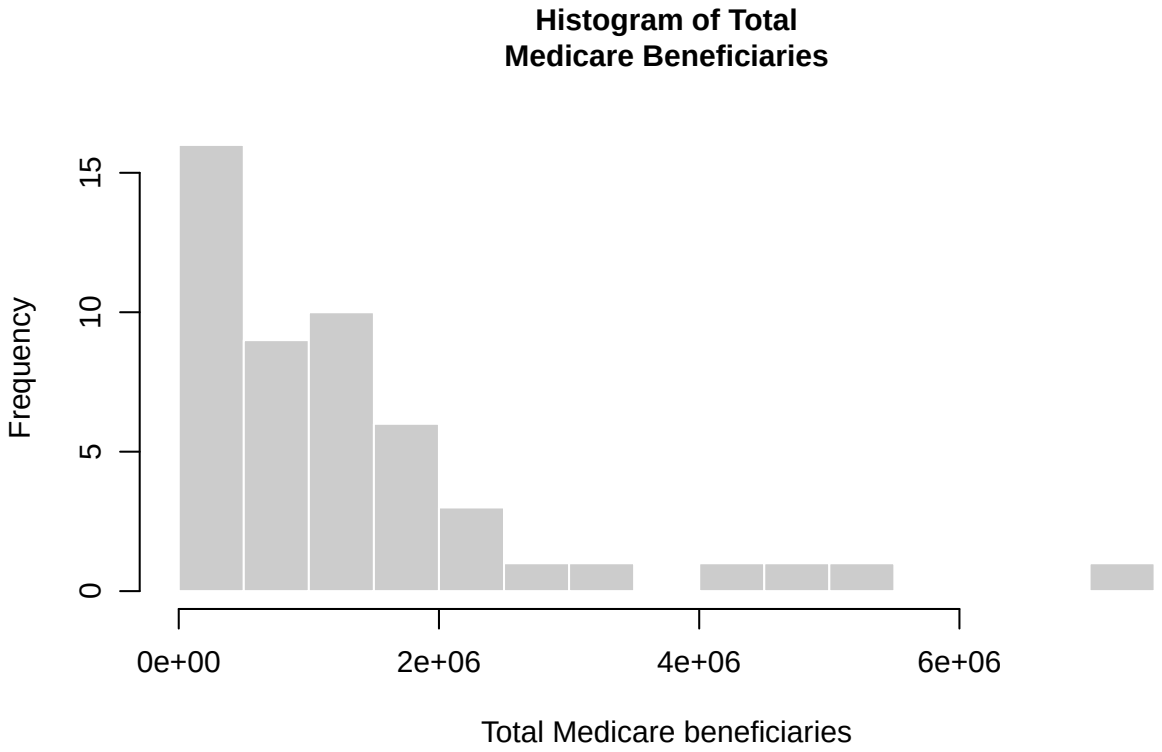
Appendix B: Data Rows

`head()` of the cleaned analysis data.

State	Beneficiaries	Avg_Age	HCC_Score	Pct_Male	Median_Income	Political_Affiliation	Education_Q	Physician_Q	Population_Q
AK	120,280	36.5	0.82	0.4918	105,752	Republican	Q2	Q3	Q1 Low
AL	1,151,244	39.6	1.02	0.4535	79,940	Republican	Q1 Low	Q1 Low	Q3
AR	699,823	38.9	1.00	0.4531	74,179	Republican	Q1 Low	Q1 Low	Q2
AZ	1,528,056	39.3	0.92	0.4635	93,553	Republican	Q2	Q2	Q3
CA	7,080,940	38.2	1.04	0.4567	109,349	Democrat	Q3	Q3	Q4 High
CO	1,055,744	37.9	0.89	0.4708	115,335	Swing	Q4 High	Q3	Q3

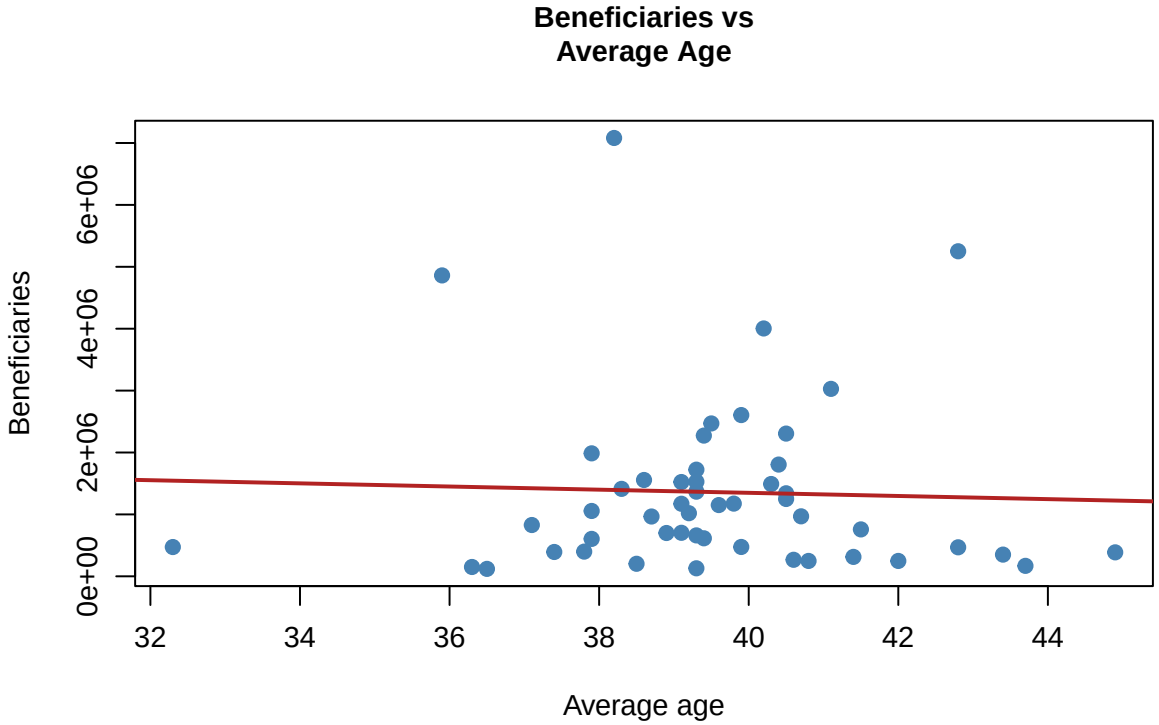
Appendix C: Exploratory Data Analysis

Response Variable Plot

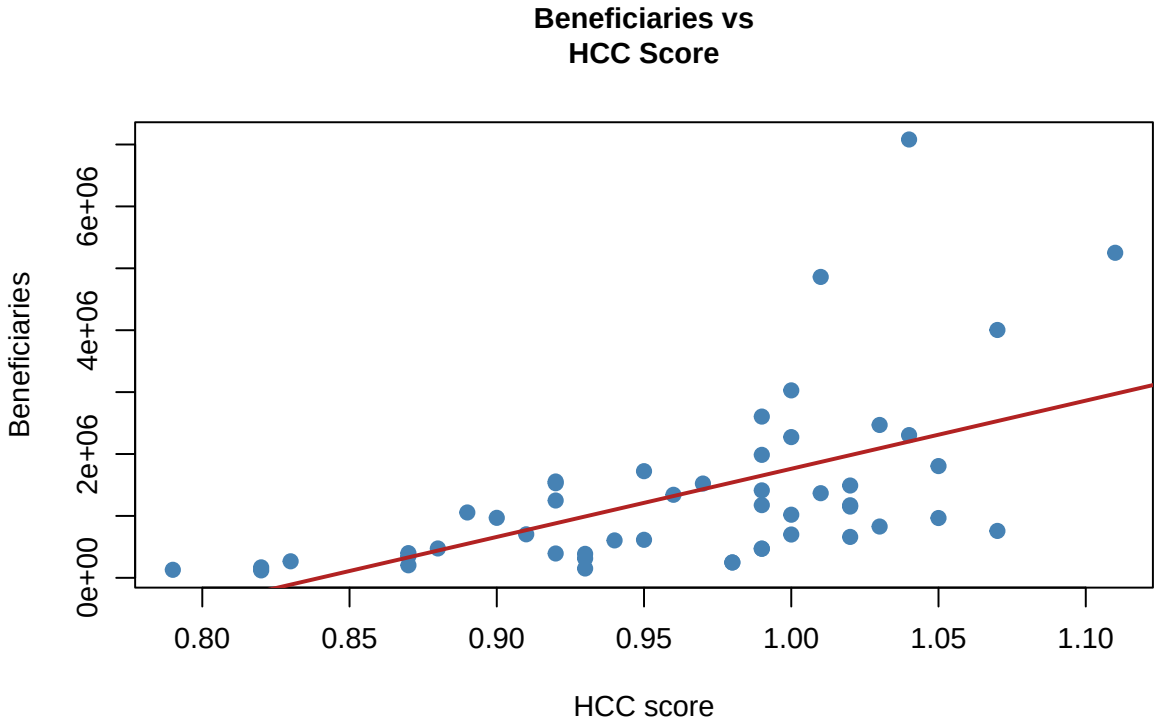


This histogram shows a strong right-skew, with most states clustered at lower beneficiary counts and a small number of very large states extending the right tail. This shape supports considering `log(Y)` in later regression work.

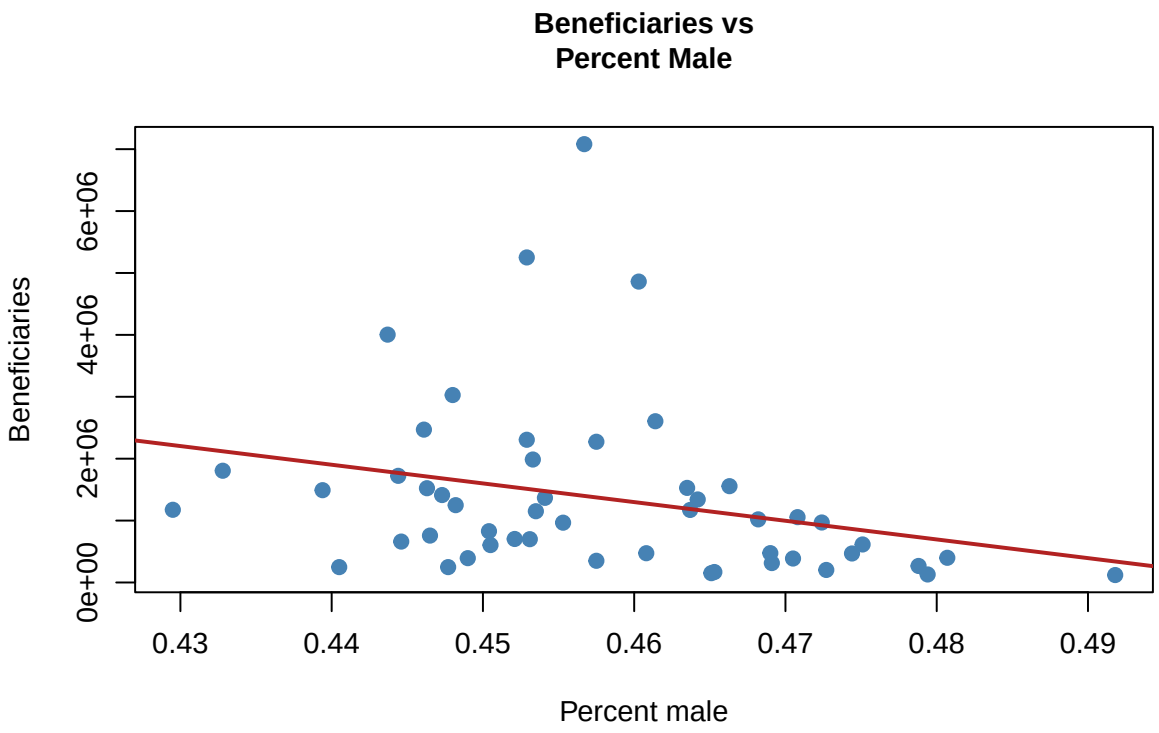
Quantitative Predictor Plots



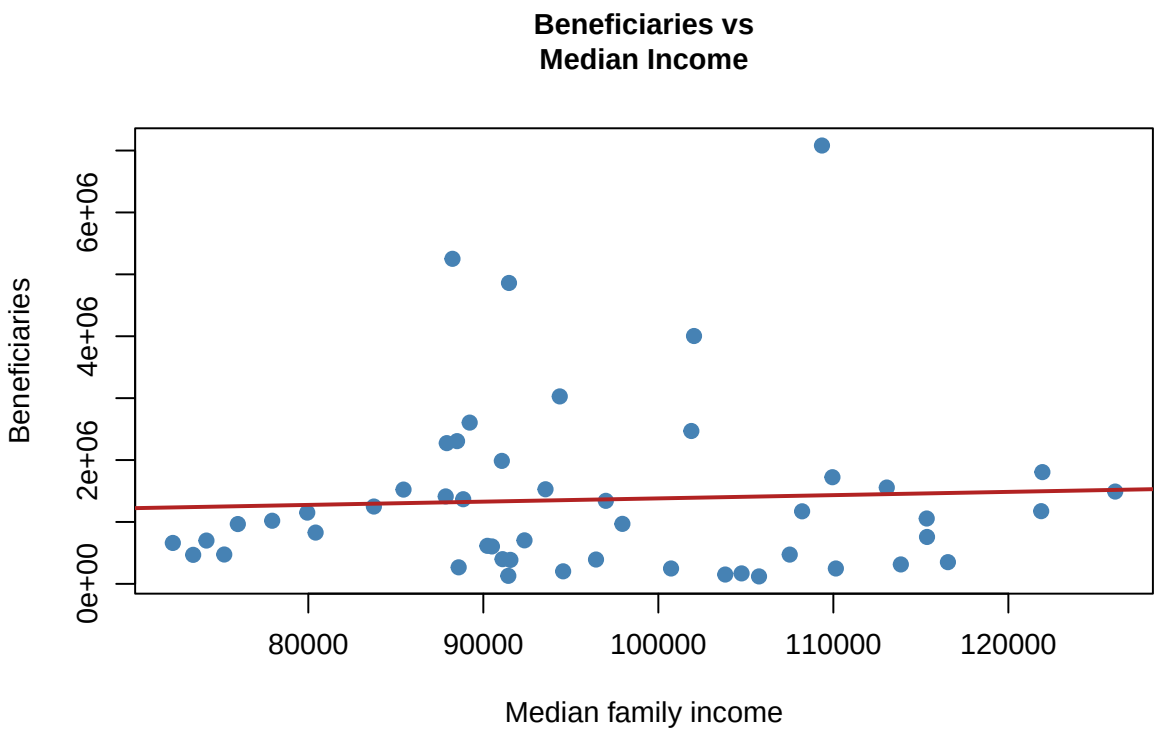
This scatterplot shows only a weak overall linear relationship between average age and beneficiary totals, which matches the small raw correlation of -0.038. The broad vertical spread suggests age may matter more through interactions than as a stand-alone predictor.



This plot shows the clearest positive quantitative pattern in the EDA. States with higher HCC scores tend to have larger beneficiary totals, consistent with the positive raw-response correlation of 0.562.

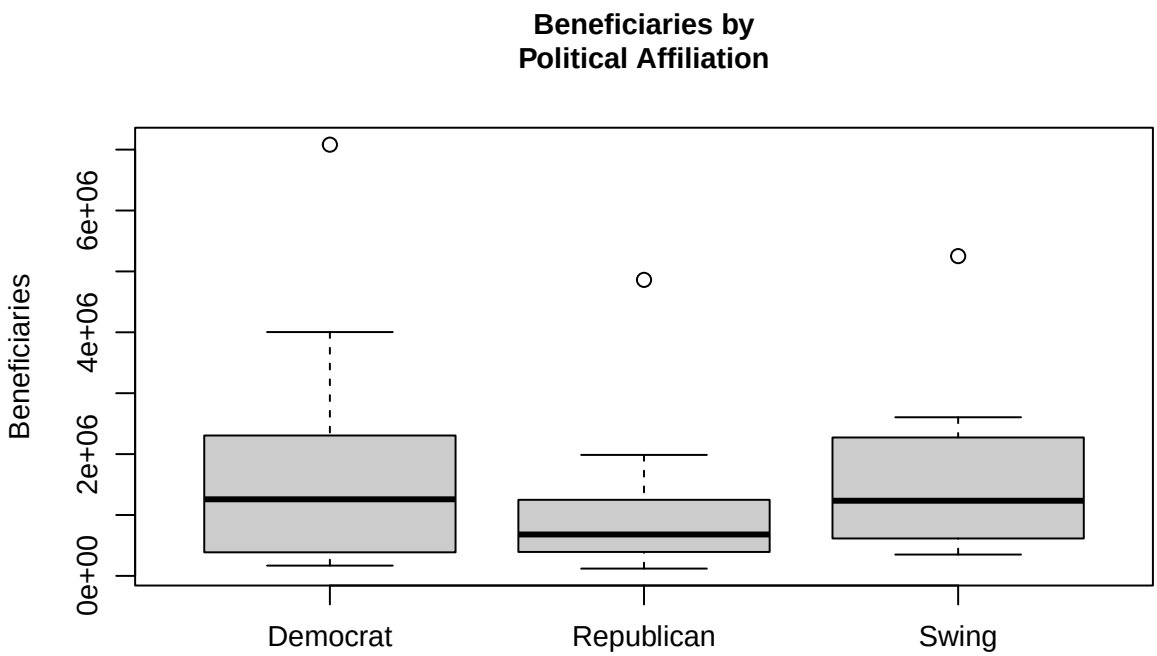


This scatterplot suggests a negative association between percent male and beneficiary totals. The downward fitted line agrees with the negative correlation of -0.28, although there is still substantial variation around the trend.



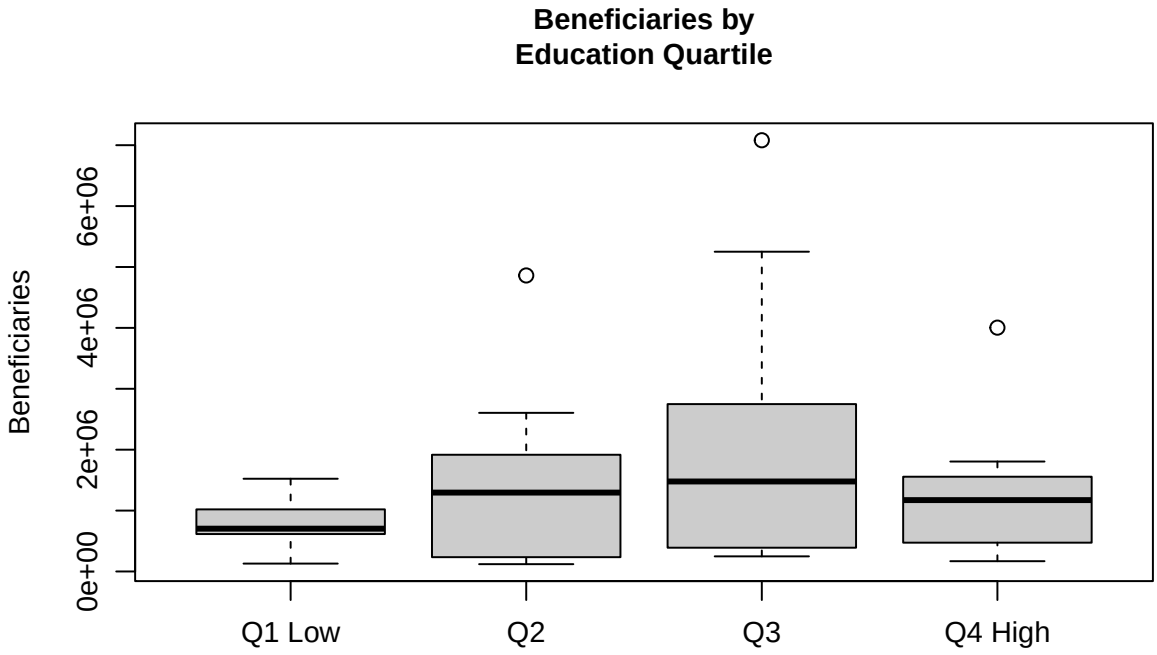
This plot shows a negligible linear relationship between median family income and beneficiary totals. The nearly flat fitted line is consistent with the very small correlation of 0.051.

Qualitative Predictor Plots

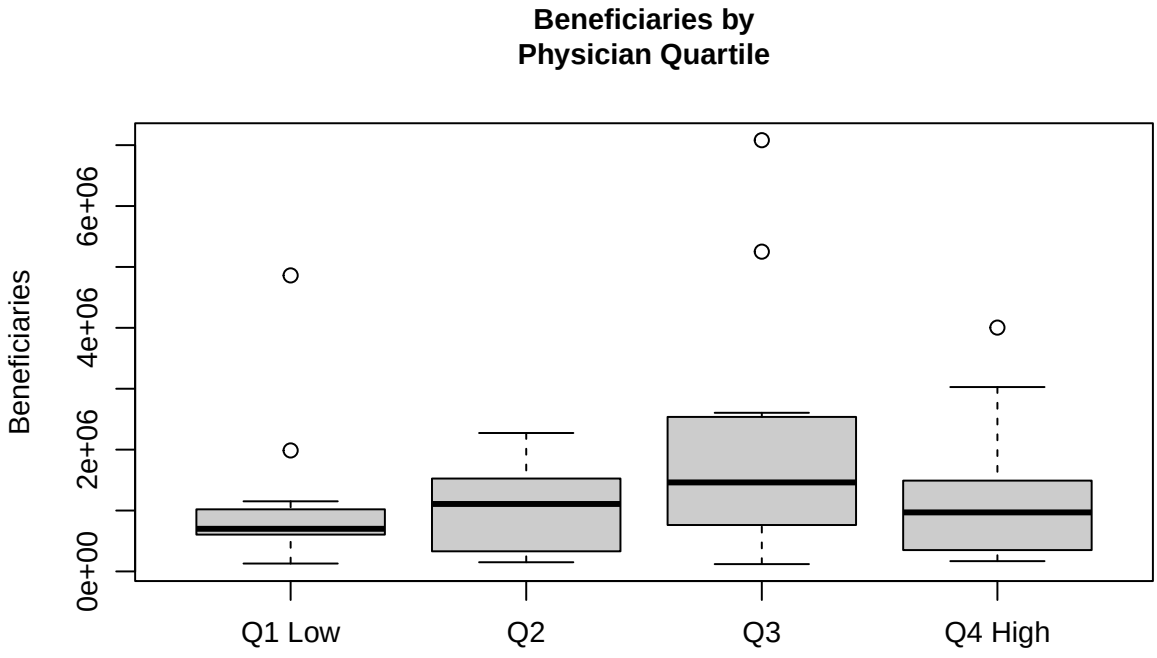


This boxplot compares beneficiary totals across political-affiliation groups. Democratic and

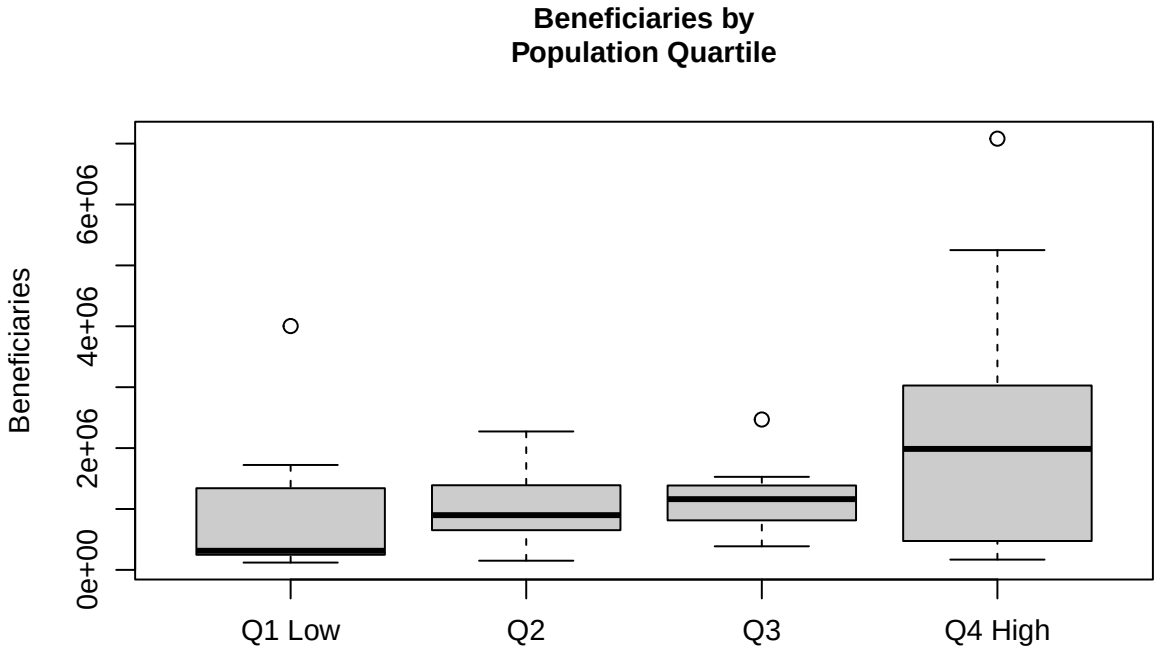
swing states have higher average totals than Republican states in this sample, although there is still overlap across the three groups.



This plot suggests some differences across education quartiles, but the pattern is not perfectly monotone. Mean beneficiary totals rise from 819,413 in the lowest quartile to 2,083,766 in Q3 before dropping in Q4.

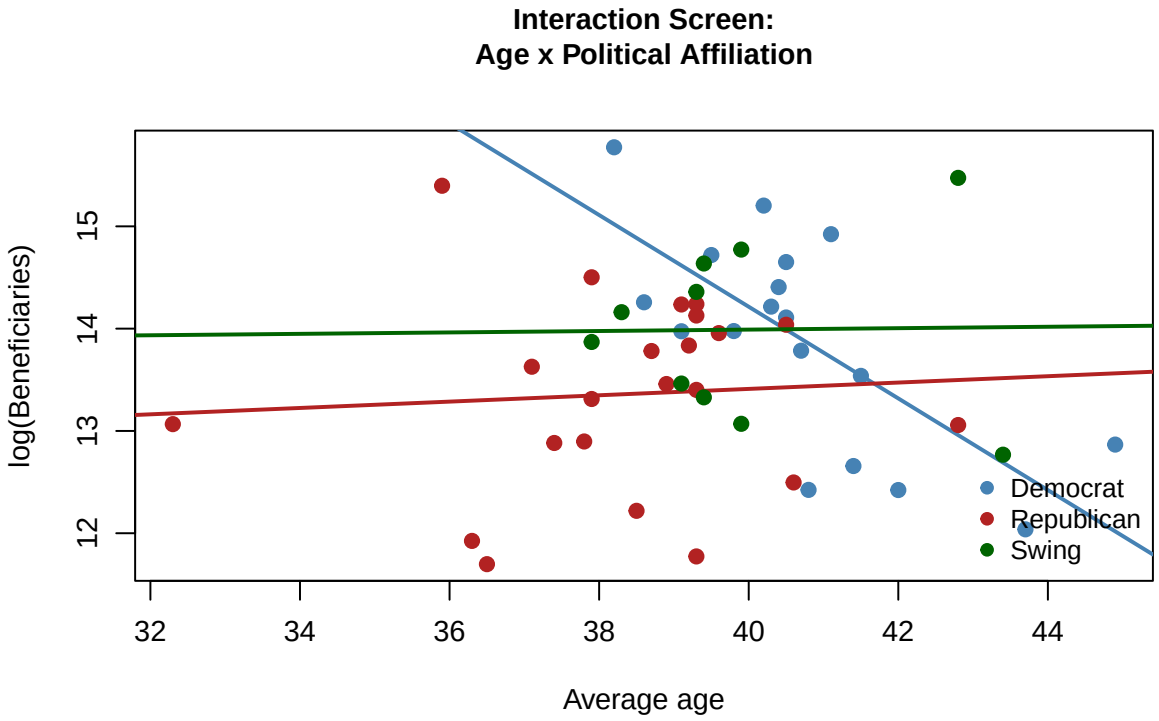


This boxplot indicates that physician availability does not increase in a simple step-by-step way with beneficiary totals. The group means vary across quartiles, but the ordering is uneven, so regression adjustment will be more informative than the unadjusted comparison alone.

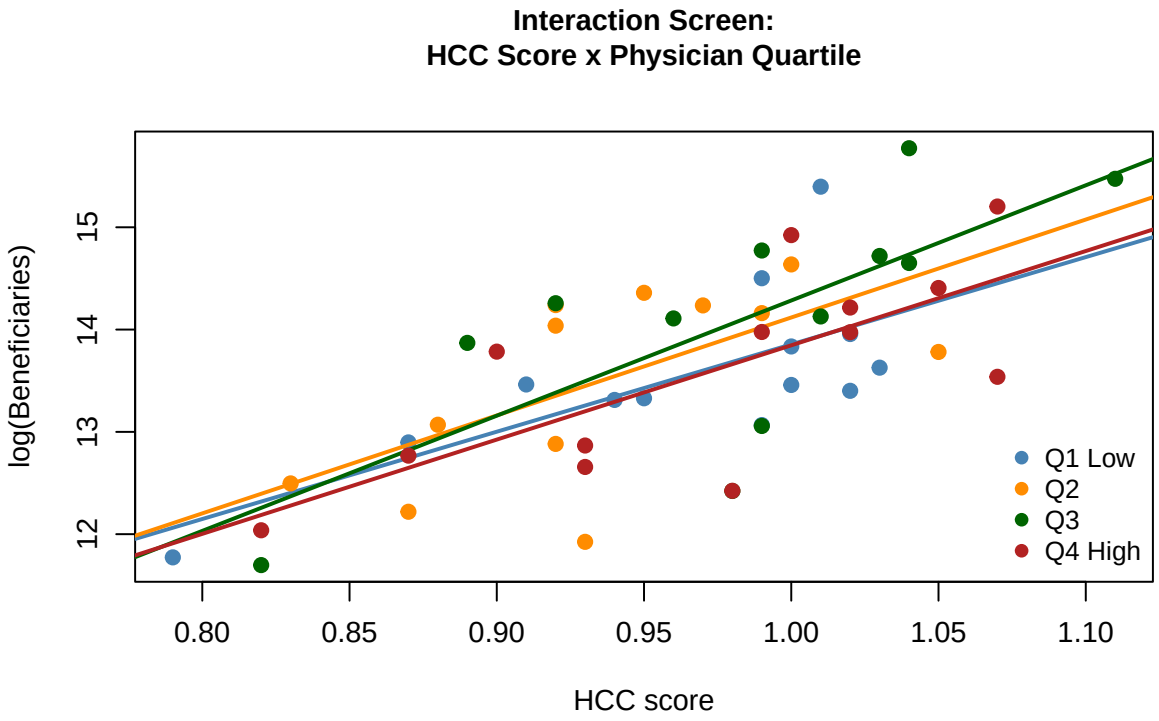


This plot shows the strongest qualitative separation in the appendix. States in higher population quartiles generally have larger beneficiary totals, with the highest-quartile mean of 2,360,061 compared with 860,492 in the lowest quartile.

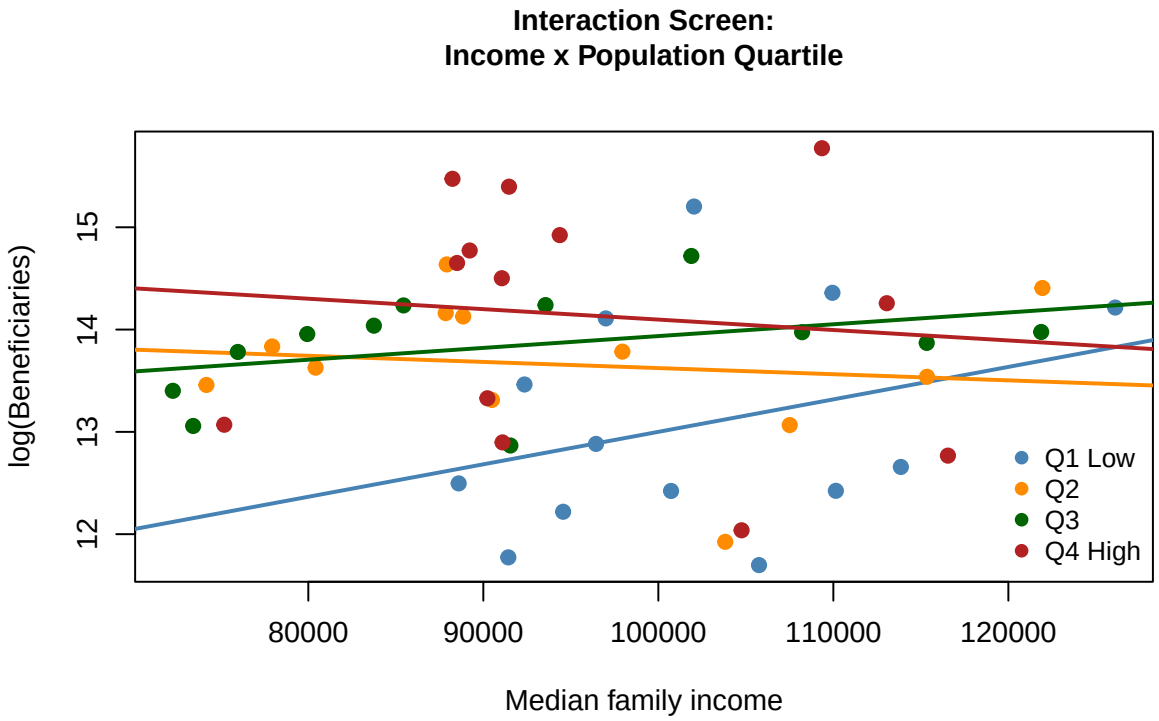
Interaction Plots



This interaction plot reveals that the relationship between a state's average Medicare age and its total number of beneficiaries depends heavily on its political affiliation. While Republican and Swing states exhibit nearly parallel, relatively flat trend lines indicating that average age has little impact on total enrollment in those regions, Democratic states display a steep downward slope. This distinct difference in slopes suggests a strong interaction effect, meaning our final model should account for political affiliation rather than assuming a uniform relationship between age and beneficiaries across all states.



This plot examines whether physician availability modifies the relationship between a state's HCC score and its total beneficiaries. Because the trend lines for all four quartiles are heavily overlapping and nearly parallel, there is no visual evidence of an interaction effect. This suggests that the relationship between a state's medical risk and its beneficiary count remains consistent regardless of the state's physician rate, thus indicating no interaction amongst these two explanatory variables.



This plot examines whether a state's population size modifies the relationship between its median income and total number of beneficiaries. Because the trend lines for the four population quartiles are roughly parallel, there is no strong visual evidence of an interaction effect altering the slopes. However, it is seen that Q1 and Q3 of population sizes have flat/slightly positive slopes, while Q2 and Q4 are slightly negative, but these differences are not major enough, subjectively, to indicate any major interaction occurring.

Appendix E: References

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Note: Some of the vocabulary, grammar, and other wording in this report was refined using ChatGPT to give the most accurate data descriptions and analysis